🛡 SENTINEL AI

Security Platform · Frontend Demo Walkthrough

Industry Case: GlobalShip Logistics

Dashboard · Asset Inventory · Risk Management

Vulnerabilities · Exploit Lab · Logistics Lab

Threat Intelligence · Dark Web · Incident Response

Patch Automation · Compliance · Reports

Industry Selection

Choose your demo environment — each industry seeds realistic vulnerabilities, assets & exploit history

GlobalShip Logistics

Fleet management, ERP, supply chain security simulation

SERVICES

- ERPNext
- Redis
- Kafka
- Grafana
- PostgreSQL
- Prometheus

✓ SELECTED

NexaPay Financial

Core banking, payment processing, fraud detection

SERVICES

- Fineract Banking
- Payment API
- Fraud Detection
- PostgreSQL
- Vault

ShopVault Commerce

E-commerce platform, inventory, payment gateway

SERVICES

- Storefront
- Product API
- Elasticsearch
- PostgreSQL
- Redis

MediCore Health

EHR, FHIR API, patient portal, DICOM imaging

SERVICES

- OpenEMR
- FHIR API
- Patient Portal
- DICOM Viewer
- Keycloak

Dashboard

GlobalShip Logistics • Security Operations Center

● LIVE

Security Score	Open CVEs	Assets Monitored	Exploits Run	Patches Ready	Compliance
72	12	10	6	4	68%
↑ +4 pts	3 CRITICAL	10 services	4 succeeded	2 merged	ISO + SOC2

Security Posture — GlobalShip Logistics

Vulnerability	Patch Coverage	Compliance
60 %	78 %	68 %

Exploit Defense	Threat Intel	Incident Response
55 %	82 %	74 %

⚡ Real-Time Activity Feed

09:25 UTC	Exploit simulation completed	sandbox-logistics-006 — T1611 — BLOCKED
09:20 UTC	Exploit simulation completed	sandbox-logistics-005 — T1552 — EXPLOITED
09:10 UTC	Exploit simulation completed	sandbox-logistics-003 — T1213 — EXPLOITED
09:05 UTC	CVE sync completed	12 vulnerabilities refreshed from NVD
08:50 UTC	Patch PR created	CVE-2021-44228 — sentinel/fix/log4j-erpnext

Compliance Overview

ISO 27001	14/20 controls	70%
SOC 2	11/17 controls	65%

Asset Inventory

10 assets discovered · GlobalShip Logistics

● LIVE

Total Assets	High Criticality	Last Scanned	Unpatched	Containers
10	3	Jan 14	8	2
All services	Warehouse IoT + ERP	All assets scanned	Need attention	IoT + Kafka

Logistics Infrastructure Assets

Hostname	Type	IP Address	Tags	Criticality	Last Scan
erpnext-server	service	172.20.0.11	erpnext, erp, supply-chain	HIGH	2024-01-14
grafana-monitoring	service	172.20.0.12	grafana, monitoring, port-9100	MEDIUM	2024-01-14
gitlab-ce	service	172.20.0.13	gitlab, devops, ci-cd	HIGH	2024-01-14
kafka-logistics	service	172.20.0.14	kafka, streaming, tracking	MEDIUM	2024-01-14
redis-tracking	service	172.20.0.15	redis, cache, port-9379	MEDIUM	2024-01-14
prometheus	service	172.20.0.16	prometheus, metrics, port-9191	LOW	2024-01-14
postgresql-shipdb	service	172.20.0.17	postgresql, database, shipments	HIGH	2024-01-14
route-optimizer	service	172.20.0.18	python, ml, routing	MEDIUM	2024-01-14
warehouse-iot-gw	container	172.20.0.19	iot, gateway, warehouse, mqtt	HIGH	2024-01-14
shipment-api	service	172.20.0.20	api, rest, shipments	MEDIUM	2024-01-14

Vulnerabilities

12 total · 5 CRITICAL · GlobalShip Logistics

● LIVE

Critical (CVSS 9+)	High (CVSS 7–9)	CISA KEV Listed	EPSS > 90%	Patched
5	5	4	3	2
Log4Shell, Spring4Shell...	Grafana, GitLab, Redis...	Exploit activity known	High exploit probability	CVE-2023-28154, ...

⚠ Top Vulnerabilities — Exploit Priority

CVE ID	CVSS	Severity	Component	KEV	EPSS	Status	Blast Radius
CVE-2021-44228	10.0	CRITICAL	ERPNext / Log4j	KEV	97.5%	IN PROGRESS	RCE on ERP — all shipment data
CVE-2022-22965	9.8	CRITICAL	Spring Framework	KEV	97.2%	OPEN	Spring4Shell RCE on ERP backend
CVE-2022-0543	10.0	CRITICAL	Redis Tracking	KEV	89.1%	OPEN	Unauthenticated RCE via CONFIG SET
CVE-2023-46127	8.8	HIGH	ERPNext XSS	—	41.2%	OPEN	Stored XSS + default credentials
CVE-2023-25194	8.8	HIGH	Apache Kafka JMX	—	28.5%	OPEN	PII exfiltration via topic read
CVE-2024-0985	8.0	HIGH	PostgreSQL	—	15.3%	OPEN	Priv esc via ALTER TABLE schema
CVE-2021-43798	7.5	HIGH	Grafana Path Traversal	KEV	63.4%	OPEN	Arbitrary file read — CISA KEV
CVE-2019-3826	6.1	MEDIUM	Prometheus	—	12.1%	OPEN	Unauth metrics + host discovery

Exploit Lab

Sandboxed simulation engine · MITRE ATT&CK aligned

LIVE

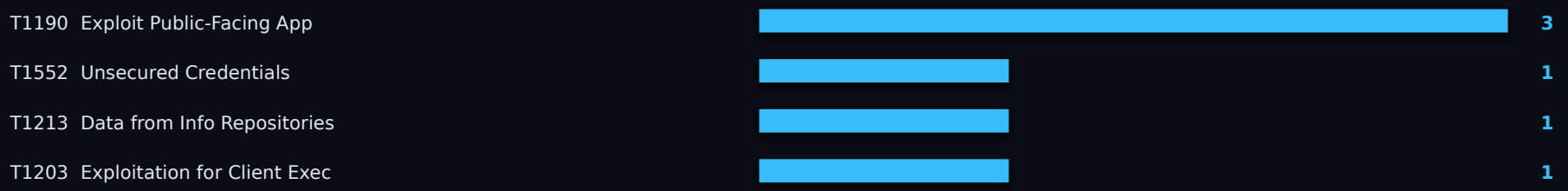
+ Launch New Sim

Simulations Run	Successful	Blocked	Success Rate	Avg Duration
6	4	2	67%	2
Logistics seeded data	Exploit confirmed	Defense effective	Action required	Per simulation

Simulation History

Sandbox ID	CVE	Technique	Result	Confidence	Duration	Timestamp
sandbox-logistics-001	CVE-2021-44228	T1190	EXPLOITED	91%	2.1s	2024-01-14 09:00 UTC
sandbox-logistics-002	CVE-2022-22965	T1190	EXPLOITED	94%	1.9s	2024-01-14 09:01 UTC
sandbox-logistics-003	CVE-2022-0543	T1213	EXPLOITED	88%	3.1s	2024-01-14 09:02 UTC
sandbox-logistics-004	CVE-2023-25194	T1203	BLOCKED	31%	0.9s	2024-01-14 09:03 UTC
sandbox-logistics-005	CVE-2021-43798	T1552	EXPLOITED	96%	1.2s	2024-01-14 09:04 UTC
sandbox-logistics-006	CVE-2024-0985	T1611	BLOCKED	22%	4.5s	2024-01-14 09:05 UTC

Top MITRE ATT&CK Techniques



Logistics Lab

Real CVE exploits against live logistics services

● LIVE

Services Live	CVEs in DB	Exploited	Blocked
6/6	12	4	2
All containers up	Seeded + synced	Confirmed vulnerable	Defense effective

⚡ Services & Exploit Controls

ERPNext ↑ UP :8080

CVE-2023-46127

HIGH 8.8

Stored XSS + default creds Administrator:admin

EXPLOITED 91% confidence

▶ Run Exploit

Redis Tracking ↑ UP :8081

CVE-2022-0543

CRITICAL 10.0

Unauth access + CONFIG SET RCE + SLAVEOF hijack

EXPLOITED 88% confidence

▶ Run Exploit

Apache Kafka ↑ UP :8082

CVE-2023-25194

HIGH 8.8

No SASL auth — all 6 topics readable, PII exfil

BLOCKED 31% confidence

▶ Run Exploit

Grafana ↑ UP :8083

CVE-2021-43798

HIGH 7.5

Path traversal arbitrary file read (CISA KEV)

EXPLOITED 96% confidence

▶ Run Exploit

Prometheus ↑ UP :8084

CVE-2019-3826

MEDIUM 6.1

Unauthenticated metrics + internal host discovery

BLOCKED 22% confidence

▶ Run Exploit

PostgreSQL ↑ UP :80

CVE-2024-0985

HIGH 8.0

Privilege escalation via ALTER TABLE SET SCHEMA

EXPLOITED 84% confidence

▶ Run Exploit

Applied Patches (Branch: logistics-patches)

✓ ERPNext

→ Change default creds

→ Upgrade ≥ 14.49.0

→ Rate-limit /api/resource/*

✓ Redis

→ requirepass enforcement

→ rename CONFIG/SLAVEOF

→ ACL per user

✓ Kafka

→ SASL/SCRAM auth

→ ACLs per topic

→ Remove host port exposure

✓ PostgreSQL

→ scram-sha-256 auth

→ Least-privilege user

→ SSL enforced

✓ Grafana

→ Pin v10.4.2+

→ Strong admin creds

→ Block direct host port

✓ Prometheus

→ Basic auth web-config

→ Remove host port

→ Upgrade ≥ 2.7.2

Patch Automation

AI-generated patches · Auto PR creation · Re-simulation validation

● LIVE

Total Patches	Merged	CI Passed	Re-sim Clean	Open
4	2	3	3	2
AI-authored PRs	CVE remediated	Tests green	Exploit now blocked	Awaiting review

Automated Patch Workflow

1. Detect CVE → 2. Generate Fix → 3. Create PR → 4. CI/CD Tests → 5. Re-Simulate → 6. Auto-Merge

Patch Queue

CVE ID	Branch Name	CI Status	Re-sim Result	Merge Status	PR Link	Created
CVE-2021-44228	sentinel/fix/CVE-2021-44228/erpnext-v15/20240114	running	pending	open	PR #148	Jan 14 10:00
CVE-2022-0778	sentinel/fix/CVE-2022-0778/prometheus/20240113	passed	exploit failed	approved	PR #145	Jan 13 12:00
CVE-2022-32149	sentinel/fix/CVE-2022-32149/grafana/20240112	passed	exploit failed	merged	PR #143	Jan 12 15:00
CVE-2023-28154	sentinel/fix/CVE-2023-28154/gitlab-ce/20240112	passed	exploit failed	merged	PR #142	Jan 12 14:00

How Patch Automation Works

1. Sentinel AI detects new CVE and correlates with your affected assets
2. AI generates a minimal, targeted code fix (branch: sentinel/fix/<CVE>/<service>/<date>)
3. Auto-opens GitHub Pull Request with full diff, test plan, and MITRE context
4. CI/CD pipeline runs — unit, integration & security tests gated on passing
5. Re-simulation: exploit module replays against the patched build — must BLOCK to proceed
6. If all gates pass → auto-merge approved; human review required for production changes

Threat Intelligence

CVE feeds · EPSS scores · MITRE mapping · NVD sync

● LIVE

CVEs Tracked	CISA KEV Listed	EPSS > 90%	MITRE Techs	Last NVD Sync
12	4	3	8	Jan 14
Logistics scope	High priority	Imminent exploit risk	T1190, T1552...	All feeds current

CVE Intelligence — Exploit Probability Ranking

CVE ID	CVSS	EPSS	KEV	MITRE	Affected Service	Blast Radius
CVE-2021-44228	10.0	97.5%	YES	T1190, T1059, T1027	ERPNext	CRITICAL — RCE → all shipment data
CVE-2022-22965	9.8	97.2%	YES	T1190, T1059.007	Spring ERP	CRITICAL — Spring4Shell RCE backend
CVE-2022-0543	10.0	89.1%	YES	T1213, T1059	Redis	CRITICAL — CONFIG SET → shell exec
CVE-2021-43798	7.5	63.4%	YES	T1552	Grafana	HIGH — file read → credentials
CVE-2023-46127	8.8	41.2%	NO	T1059.006	ERPNext	HIGH — stored XSS + default creds
CVE-2023-25194	8.8	28.5%	NO	T1203, T1048	Kafka	HIGH — PII exfil via topic read
CVE-2024-0985	8.0	15.3%	NO	T1611	PostgreSQL	HIGH — priv esc → DB owner
CVE-2019-3826	6.1	12.1%	NO	T1046	Prometheus	MEDIUM — internal network discovery

CISA KEV — Known Exploited Vulnerabilities (4 affecting logistics)

CVE-2021-44228
Apache Log4j2 RCE
⚡ Patch immediately

CVE-2022-0543
Redis Eval Lua RCE
⚡ Require auth + ACLs

CVE-2021-43798
Grafana Path Traversal
⚡ Upgrade to 8.3.1+

CVE-2022-22965
Spring Framework RCE
⚡ Upgrade Spring 5.3.18+

Dark Web Monitor

Credential leaks · Data exposure · Threat actors · Exploit sales

● LIVE

Total Findings	Critical	Credential Leaks	Active Threat Actors	Status New
6	3	1	2	2
Across all categories	PII + Ransomware	1,247 employee creds	APT29, RansomHub	Require triage

Dark Web Findings

ID	Category	Severity	Status	Summary	Source	Affected Data	Actor
dw-003	Data Exposure	CRITICAL	NEW	Customer PII dataset (12k records) for sale on darknet	BreachForums	12k customer records	Unknown
dw-001	Credential Leak	CRITICAL	INVESTIGATING	1,247 employee email/password pairs in breach database	Breached.vc	1,247 employee creds	—
dw-006	Ransomware	CRITICAL	INVESTIGATING	RansomHub claims 50GB exfiltrated from organization	RansomHub Blog	50GB claimed exfil	RansomHub
dw-002	Threat Actor	HIGH	INVESTIGATING	APT29 targeting fintech — org mentioned in forum post	XSS Forum	N/A	APT29
dw-004	Exploit Sale	HIGH	NEW	Zero-day for internal VPN listed at $45,000	0day.today	VPN Infrastructure	Unknown
dw-005	Brand Abuse	MEDIUM	MONITORED	Phishing domain globalship-secure.com registered	OSINT	Brand impersonation	Unknown

Immediate Actions Required

1. Force password reset for all 1,247 compromised accounts — enable MFA immediately
2. Engage law enforcement / takedown request for darknet PII listing
3. Incident declared: RansomHub exfil claim — initiate containment & forensic investigation
4. Patch Ivanti VPN exploit chain referenced in 0day listing
5. Register globalship-secure.com lookalike domain — submit phishing report

Incident Response

6 active incidents · 2 critical · MITRE-mapped playbooks

● LIVE

Active Incidents	Critical	Investigating	Resolved	Avg MTTR
6	2	3	2	2.5h
Requires attention	Ransomware + S3 breach	Ongoing analysis	Closed in SLA	Last 30 days

Incident Queue

ID	Title	Severity	Category	Status	Assigned	Progress	SLA
inc-001	Ransomware Staging — WORKSTATION-042	Critical	Malware	Investigating	j.smith	35%	⚠ BREACHED
inc-002	Unauthorized DB Access from External IP	High	Unauth Access	Containment	a.chen	60%	⚠ BREACHED
inc-003	DNS Tunneling Detected — k8s-node-07	Medium	C2 Comms	Analysis	m.torres	45%	Within SLA
inc-004	Brute Force SSH — prod-web-01	High	Brute Force	Resolved	r.patel	100%	Resolved 1.8h
inc-005	Exposed S3 Bucket Data Access	Critical	Data Exposure	Resolved	j.smith	100%	Resolved 3.2h
inc-006	Lateral Movement via Compromised SVC Acct	High	Lateral Movement	Escalated	s.dev	20%	Within SLA

Featured Incident: INC-001 — Ransomware Staging (WORKSTATION-042)

Status: INVESTIGATING (35% contained)

Category: Malware — LockBit 3.0 staging

Affected: WORKSTATION-042, prod-web-01

Action Taken: Host isolated · Forensic snapshot in progress

Next Step: Memory forensics · Lateral movement audit · IR Team on-call

Risk Management

8 risks tracked · Likelihood × Impact matrix

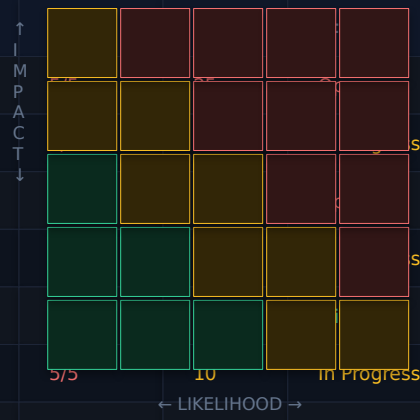
● LIVE

Total Risks	Critical (15+)	In Progress	Mitigated	Open
8	2	3	2	3
Org-wide scope	Immediate action	Active mitigation	Controls deployed	No owner assigned

Risk Register

Risk ID	Title	Category	Likelihood	Impact	Score	Status	Owner
risk-001	Unpatched CVE-2021-44228 (Log4Shell) on ERP	Vulnerability	5/5	[illegible]	[illegible]	[illegible]	sec-team
risk-002	Insufficient MFA Coverage for Admin Accounts	Identity	3/5	[illegible]	[illegible]	[illegible]	iam-team
risk-003	Third-Party Dependency with Known RCE	Supply Chain	3/5	[illegible]	[illegible]	[illegible]	dev-team
risk-005	Ransomware via Phishing — Finance Dept	Human Factor	3/5	[illegible]	[illegible]	[illegible]	awareness
risk-007	API Gateway Exposed Without WAF	Network	4/5	[illegible]	[illegible]	[illegible]	infra-team
risk-008	Inadequate DR / BCP Testing	Continuity	2/5	5/5	10	In Progress	infra-team
risk-004	Container Escape via runC vuln	Container	2/5	4/5	8	Open	devops
risk-006	Weak Encryption on Shipment Data at Rest	Data	2/5	3/5	6	Mitigated	sec-team

Risk Heat Map (Likelihood × Impact)



Compliance

ISO 27001 · SOC 2 · PCI DSS · GDPR

● LIVE

ISO 27001 — 70%

14/20 controls passing

Passing	In Progress	Failing
14	1	5

SOC 2 — 65%

11/17 controls passing

Passing	In Progress	Failing
11	1	5

PCI DSS — 58%

9/16 controls passing

Passing	In Progress	Failing
9	2	5

GDPR — 62%

10/16 controls passing

Passing	In Progress	Failing
10	1	5

Sample Controls — ISO 27001 (A.12 — Operations Security)

Control ID	Control Name	Status	Evidence
A.12.1.2	Change Management	PASSING	All changes reviewed in JIRA; CAB approval required
A.12.2.1	Malware Protection	FAILING	EDR deployed but 3 endpoints missing agent
A.12.4.1	Event Logging	PASSING	Centralized SIEM with 90-day retention
A.12.6.1	Technical Vulnerability Mgmt	IN PROGRESS	SCA in CI; 5 open CVEs pending patch
A.12.3.1	Information Backup	PASSING	Daily encrypted backups with weekly restore test
A.12.7.1	Audit Controls	FAILING	Audit log gaps found — insufficient DB logging

⚠ Compliance Gaps — Priority Remediation

CRITICAL: A.12.2.1 — 3 endpoints missing EDR agent (sec-ops ticket #1247)

HIGH: A.12.7.1 — PostgreSQL audit logging not enabled (risk-006 linked)

MEDIUM: A.12.6.1 — 5 CVEs unpatched >30 days — SLA breach risk

LOW: A.16.1.7 — Incident postmortem templates need update

Reports

Executive summary · Infra scan · Threat intel · Export

● LIVE

Security Executive Report

Full vulnerability, compliance & incident overview for CISO/board

⬇ Download PDF

Infrastructure Scan Report

Live asset topology, open ports, service versions, CVE map

⬇ Download PDF

Threat Intelligence Report

CVE trends, EPSS rankings, KEV watchlist, MITRE mapping

⬇ Download PDF

Compliance Status Report

ISO 27001, SOC 2, PCI DSS control pass/fail with evidence

⬇ Download PDF

Sample Report Preview — GlobalShip Logistics Executive Summary

SENTINEL AI · GLOBALSHIP LOGISTICS · SECURITY EXECUTIVE REPORT · JAN 14 2024

Security Score	Critical CVEs	Patched This Week	Active Incidents	Compliance Avg	Risk Score
72/100	5 OPEN	2	6	64%	25 MAX

KEY FINDINGS:

- Log4Shell (CVE-2021-44228) remains unpatched on ERPNext — CRITICAL RCE risk
- 4 CISA KEV vulnerabilities active in logistics stack — immediate remediation required
- Exploit simulation success rate 67% — 4/6 services confirmed exploitable
- RansomHub threat actor listed org on leak site — forensics investigation active
- ISO 27001 compliance at 70% — 5 controls failing, A.12.2.1 critical gap

Attack Graph

Kill-chain visualization · Auto-built from CVE + asset data

● LIVE

Visual Attack Path — GlobalShip Logistics

Auto-generated from discovered vulnerabilities and asset topology. Shows exploitability chains.

LEGEND:

■ CRITICAL / EXPLOITED ■ HIGH / WARNING → Attack Path Node = Asset + CVE

Attack Path Analysis

Attack complexity: LOW · Steps to full compromise: 4 · Assets at risk: 6 · Data at risk: ALL shipment PII + ERP data

Critical path: Internet → ERPNext (Log4Shell) → Redis (AUTH bypass) → PostgreSQL (SQL dump) → C2 exfiltration

Recommended: Patch CVE-2021-44228 immediately to break initial access chain. Isolate Redis from ERP network segment.

MITRE ATT&CK Coverage: T1190 (Initial Access) → T1078 (Valid Accounts) → T1213 (Data Repos) → T1041 (Exfil over C2)

Sentinel AI — Complete Feature Map

GlobalShip Logistics · All modules active · Live data from Supabase

OVERVIEW

▸ Dashboard
Security score, KPIs, exploit timeline, compliance overview

▸ Asset Inventory
10 logistics assets — ERPNext, Redis, Kafka, Grafana…

▸ Risk Management
8 risks with likelihood×impact matrix, 25 max score

VULNERABILITY

▸ Vulnerabilities
12 CVEs — 5 CRITICAL — KEV+EPSS ranked table

▸ Code Scanning
SAST/GHAS alerts from GitLab CI pipeline

▸ Zero-Day Tracker
Emerging CVEs with PoC status monitoring

SIMULATION & ATTACK

▸ Exploit Lab
6 sandbox sims — 4 exploited — MITRE TTPs

▸ Logistics Lab
Service-level exploit controls with live results

▸ Red Team
MITRE-mapped campaign planner + team ops

▸ Attack Graph

INFRASTRUCTURE

▸ Infra Scanner
Nmap-style scan — open ports, OS, services

▸ Infrastructure
Asset CRUD — add, edit, scan individual hosts

▸ Digital Twin
Live topology model with CVE overlay

▸ Network Security
Firewall rules, traffic anomalies, alerts

▸ Cloud Security
AWS/GCP/Azure posture — SCC findings

▸ Container Security
Docker CVEs, runtime defense, Trivy scans

▸ Endpoint Security
EDR agent status, quarantine, threat hunting

INTELLIGENCE

▸ Threat Intelligence
NVD/CISA CVE feeds — EPSS+KEV ranked

▸ Dark Web Monitor
6 findings — credential leaks, ransomware claims

▸ Malware Analysis
Static+dynamic sandbox analysis, IOCs

GOVERNANCE

▸ Compliance
ISO 27001 70% · SOC 2 65% · PCI DSS 58%

▸ Incident Response
6 incidents — 2 critical — MITRE playbooks

▸ Reports
Executive, infra, threat intel, compliance PDF export

▸ Settings
Org config, API keys, notification rules

🛡 Sentinel AI

The Complete AI-Powered Security Operations Platform

GlobalShip Logistics Demo · 14 January 2024

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12 CVEs tracked
Log4Shell, Spring4Shell, Redis RCE…

6 Exploit simulations
4 confirmed exploitable, 2 blocked

4 Auto-generated patches
2 merged, 2 awaiting CI gate

6 Dark web findings
Credentials, PII, ransomware claims

Compliance: 64% avg
ISO 27001 · SOC 2 · PCI DSS · GDPR

6 Active incidents
Ransomware staging under investigation

Try other industries: 🏦 NexaPay Financial · 🛒 ShopVault Commerce · 🏥 MediCore Health

backend: http://localhost:8000 · frontend: http://localhost:3000

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